

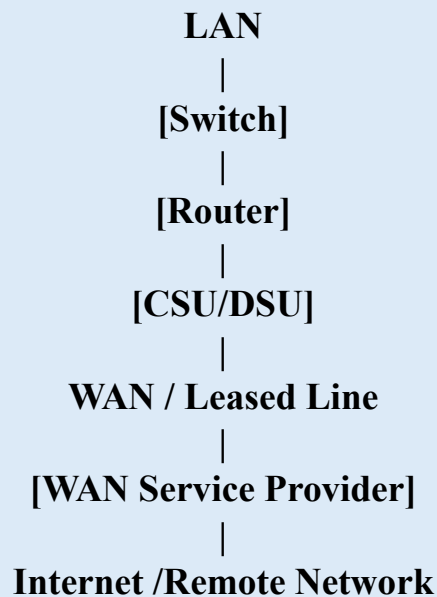
INRO TO WAN TECHNOLOGY

1. Create a table comparing at least three types of WAN connections (such as leased line, DSL, and MPLS) with columns for speed, cost, and typical use cases.

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WAN Connection	Speed	Cost	Typical Use Cases
Leased Line	High and dedicated	High	Businesses needing a dedicated, reliable connection between offices
DSL	Low to Medium	Low	Small offices, homes, and basic internet access
MPLS	Medium to High	High	Large companies connecting multiple branch offices securely
Fiber WAN	Very High	Medium to High	Enterprises, data centers, and organizations needing high-speed connectivity

2. Draw a labeled diagram (on paper or using any drawing tool) showing how a CSU/DSU connects a router to a WAN service provider, and explain the role of each device in 2-3 sentences. ○



Role of Each Device

1. Router:

The router connects the local network to the WAN. It forwards data packets between the LAN and the remote network.

2. CSU/DSU:

The CSU/DSU connects the router's digital data interface to the service provider's WAN circuit. It converts and manages the digital signals so they can travel over the leased WAN line.

3. WAN Service Provider:

The WAN service provider provides the communication link between different locations. It carries data from the customer's router to the remote network or another branch.

3. List three real-world examples of popular apps or services (like IRCTC, WhatsApp, or Zomato) that rely on WAN connections, and briefly describe how WAN enables their functionality. ○

Real-World Examples of WAN Connections

- IRCTC: WAN connects users across India to IRCTC's central servers, allowing them to search trains, book tickets, and check PNR status from different locations.
- WhatsApp: WAN allows users in different cities or countries to exchange messages, photos, videos, and make voice/video calls through internet-connected servers.
- Zomato: WAN connects customers, restaurants, delivery partners, and Zomato's servers, allowing users to order food, track deliveries, and receive real-time updates.

4. Research online and find an image or datasheet of a physical WAN aggregator device used in enterprise networks. Paste the image or link, and write two key features of that device. ○



Physical WAN Aggregator Device: Cisco Catalyst 8300

Official Datasheet: [Cisco Catalyst 8300 Series Edge Platforms Datasheet](#)

Two Key Features:

1. High-speed WAN connectivity: It provides multiple Ethernet WAN/LAN interfaces, with supported port speeds ranging from 10 Mbps up to 10 Gbps depending on the model.
2. Integrated SD-WAN and security: It supports Cisco Catalyst SD-WAN and hardware-accelerated IPsec VPN, allowing enterprises to securely connect branch offices over WAN networks.